

Oriolowo Mustapha

Software Engineer (Backend / Full-Stack)

Abeokuta, Ogun State, Nigeria | +234 703 160 2720 | oriolowomustapha@gmail.com
Portfolio: oriolowomustapha.me | GitHub: <https://github.com/Oriolowo-Mustapha> |
LinkedIn: <https://www.linkedin.com/in/oriolowo-mustapha-29b769394/>

PROFESSIONAL SUMMARY

Backend-focused Software Engineer with strong full-stack experience, building secure, scalable web applications and REST APIs with TypeScript/Node.js (Express) and C# / ASP.NET Core. Experienced in designing clean, maintainable architectures, integrating AI/LLM services (OpenAI, Gemini) for real-world workflows, and implementing background processing with Redis + BullMQ. Skilled in MongoDB/PostgreSQL, JWT/OAuth authentication, and cloud media pipelines (Multer + Cloudinary). Comfortable collaborating in agile teams, documenting APIs (Swagger/OpenAPI), and delivering production-ready features end-to-end.

TECHNICAL SKILLS

- Languages: TypeScript, JavaScript, C#, SQL, Python, HTML5, CSS3
- Frontend: React, Tailwind CSS, Vite, React Router, TanStack Query (React Query), Zustand, Axios, React Hook Form
- Backend: Node.js, Express.js, ASP.NET Core Web API, ASP.NET Core MVC/Razor, Entity Framework Core, Mongoose
- Databases: MongoDB, PostgreSQL, MySQL
- Auth & Security: JWT, Refresh Tokens, Role-Based Access Control (RBAC), Google OAuth (Passport)
- AI / LLM Integrations: OpenAI SDK, Google Generative AI (Gemini), multi-model fallback patterns, structured JSON outputs (Zod), JSON repair
- Background Jobs & Infra: BullMQ, Redis
- DevOps & Tools: Swagger/OpenAPI, Postman, ESLint
- Cloud & Media: Cloudinary, Multer
- Logging & Monitoring: Winston
- Architecture & Patterns: Clean Architecture, CQRS (MediatR), Service/Repository Patterns, REST API Design, Data Modeling

PROJECTS

AgroGuardian_AI — Backend Engineer (Contributor) **(TypeScript / Node.js)**

Live: <https://agroguardianai.pxxl.click> | GitHub:

https://github.com/Adeyemiadigun/AgroGuardian_AI

Tech Stack: TypeScript, Node.js, Express, MongoDB (Mongoose), BullMQ + Redis, JWT, Google OAuth (Passport), Zod, Multer, Cloudinary, OpenAI SDK, Gemini (@google/generative-ai), Winston

- Contributed to an AI + analytics backend for climate-smart agriculture and regenerative farming, delivering decision support via climate risk analytics and sustainability/carbon-credit tracking.
- Implemented/extended a carbon credit module that estimates CO2e sequestration using area carbon factor crop multiplier duration, adjusting outputs by soil type, climate zone (latitude-derived), crop variety, and additionality eligibility rules.
- Contributed to a climate risk engine that produces context-aware risk scores (drought, flood, heat, pest, disease) by combining weather signals with farm context (e.g., soil drainage and irrigation method).
- Built/maintained vision-based verification workflows using Gemini/OpenAI to validate uploaded evidence photos for sustainable practices; stored media via Cloudinary.
- Supported multi-model AI integration with fallback/self-healing behavior; enforced structured outputs with Zod schemas and JSON repair.
- Helped offload heavy processing (AI analysis, weather sync) to BullMQ workers backed by Redis to keep API responses responsive.

Gatepass Code Generator System — Backend Engineer (.NET)

GitHub: <https://github.com/Oriolowo-Mustapha/Gatepass-Code-Generator-System>

Tech Stack: .NET 10, ASP.NET Core Web API, PostgreSQL, EF Core 10, JWT + Refresh Tokens, MediatR (CQRS), FluentValidation, Swagger, QRCoder, BCrypt.Net, MailKit

- Built a secure, role-based Gatepass Management REST API using Clean Architecture and CQRS patterns.
- Implemented JWT authentication with refresh tokens, RBAC (Administrator, Security), and forgot/reset password flows via email.
- Delivered gatepass request workflows with QR code generation and verification; implemented visitor check-in/check-out tracking, notifications, audit logs, and reporting endpoints.

VeriLens — Backend Engineer (TypeScript)

Live: <https://verilens.pxxl.click> | GitHub:

<https://github.com/Oriolowo-Mustapha/VeriLens>

Tech Stack: Node.js, Express, TypeScript, MongoDB; Frontend: HTML, Tailwind CSS,

Vanilla JS

- Built an AI-powered fake news detection platform that analyzes articles and images and classifies results as Fake, Real, or Suspicious.
- Implemented AI news verification, image claim verification, credibility scoring, real-time analysis, user auth/history, and admin management.
- Structured backend modules (controllers, middleware, models, routes, services) and documented endpoints (API_DOC).

Agroguardian Frontend — Frontend Engineer

Live: <https://agroguardian-frontend.vercel.app> | GitHub:

<https://github.com/Oriolowo-Mustapha/Agroguardian---Frontend>

Tech Stack: React (SPA), Vite, JavaScript, Tailwind CSS, React Router, Axios, TanStack Query, Zustand, React Hook Form, Zod, Framer Motion, ESLint, Vercel

- Built a React single-page application for farm weather risk insights, designed to integrate with the AgroGuardian_AI backend.
- Implemented a manual “weather sync” flow that triggers backend synchronization, shows loading/success/error states, and refreshes the UI after completion.
- Developed a 7-day Weather Risk analysis experience showing drought, flood, heat stress, pest, and disease risk as cards with level badges, 0–100 scores, progress bars, and contributing-factor explanations.
- Added clear visual semantics (green/amber/red thresholds) to make risk severity interpretable at a glance.
- Displayed farm context (soil type and irrigation method) to help users understand why risk results vary by farm conditions.
- Designed the UI to be backward-compatible with both simple risk levels and optional detailed risk objects (score + factors) when available from the backend.

Skill-Matrix — Full-Stack Engineer (ASP.NET Core)

Live: <https://skill-qtxf.onrender.com/> | GitHub:

<https://github.com/Oriolowo-Mustapha/Skill-Matrix>

Tech Stack: C#, ASP.NET Core MVC/Razor, HTML/CSS/JavaScript, Entity Framework (migrations/entities/DTOs), Docker, Git

- Built a web-based employee skills mapping and team capability management application for team leads, engineering managers, and HR/operations to maintain an accurate “skills inventory” across an organization.
- Implemented employee profiles with identity and professional details (role, contact info, certifications, summaries) to support talent visibility and internal mobility.
- Designed skill tracking per employee with proficiency levels (not just present/absent), enabling clearer gap analysis and professional development planning.
- Developed a manager dashboard (“skills matrix”) to visualize skill coverage and strengths across teams and reduce single points of failure.
- Added team management features to create teams and assign employees,

- improving staffing and project allocation workflows.
- Implemented search/sort capabilities to quickly find people by skill and proficiency (e.g., locating strong React developers).
- Structured skills into categories/domains (e.g., technical, soft skills, digital competence, business/entrepreneurship) for consistent organization-wide reporting.
- Containerized the application with Docker and deployed to Render for reproducible builds and consistent hosting.

AjoCore (DevCareer x Nomba Hackathon) — Full-Stack Engineer

Live Demo: <https://ajo-core-frontend-eta.vercel.app>

GitHub: <https://github.com/Oriolowo-Mustapha/AjoCore---Backend> |

<https://github.com/Adeyemiadigun/AjoCore--Frontend>

Tech Stack: React 19, TypeScript, .NET 10, ASP.NET Core Web API, PostgreSQL, EF Core 10, JWT, MediatR (CQRS), FluentValidation, Hangfire, Nomba API, Brevo

- Engineered a full-stack cooperative savings platform (Ajo/Esusu) for the DevCareer and Nomba Hackathon, digitizing group-based financial cycles and ledgers.
- Built a secure, role-based REST API using Clean Architecture and CQRS patterns, featuring JWT authentication and three-tier RBAC (Trader, Cooperative Admin, System Admin).
- Integrated the Nomba API for dynamic virtual account provisioning and processed real-time inbound payments via cryptographically verified webhooks.
- Delivered an automated rotational payout engine and leveraged Hangfire background jobs for liquidation sweeps, failed transaction reversals, and email reminders.

EDUCATION

Federal University of Agriculture, Abeokuta (FUNAAB) — B.Sc. Computer Science

Expected Graduation: August 2028

Relevant Coursework: Data Structures & Algorithms, Software Engineering, OOP, DBMS, Web Development

•